

- 6 The curve C has Cartesian equation $x^2 + xy + y^2 = a$, where a is a positive constant.

(a) Show that the polar equation of C is $r^2 = \frac{2a}{2 + \sin 2\theta}$. [3]

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(b) Sketch the part of C for $0 \leq \theta \leq \frac{1}{4}\pi$. [2]

(c) It is given that $\sin 2\theta$ may be expressed as $\frac{2 \tan \theta}{1 + \tan^2 \theta}$. Use this result to show that the area of R is

$$\frac{1}{2}a \int_0^{\frac{1}{4}\pi} \frac{1 + \tan^2 \theta}{1 + \tan \theta + \tan^2 \theta} d\theta$$

and use the substitution $t = \tan \theta$ to find the exact value of this area. [8]

[illegible]